INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

Domaine Scientifique de la DOUA
20, Avenue Albert Einstein - 69100 Villeurbanne

Scheduling
Scheduling

IDENTIFICATION

CODE : GI-4-S1-EC-ORD
ECTS : 2

HOURS

Cours :	0h
TD :	8h
TP :	14h
Projet :	0h
Evaluation :	2h
Face à face pédagogique :	24h
Travail personnel :	12h
Total :	40h

ASSESMENT METHOD

IE (written test)
ES (situational assessment)

TEACHING AIDS

Handout for the lecture-tutorials part with exercises.
Project description.
Access to Opcenter APS software package.

TEACHING LANGUAGE

French
English

CONTACT

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AIMS

This course belongs to teaching unit Pilotage des opérations industrielles (GI-4-S1-UE-POPI) and contributes to the following skills :
A1 Analyze a system (real or virtual) or a problem (Level 3)
A2 Operate a model of a real or virtual system (Level 3)
A3 Implement an experimental approach (Level 2)
C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
C2 Modeling and designing an information, decision and production system, of goods and services (Level 3)
C3 Evaluating, prototyping and simulating a system (Level 3)
C5 Managing a production system and react to malfunctions (Level 3)
C6 Selecting appropriate production tools, integrating them into an environment and configuring them, and setting up a production system (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :
- Production scheduling techniques and methods: single machine, parallel machines, flowshop, jobshop, hybrid configurations. (A1, A2, A3, C1, C2, C3, C5, C6)

To be able to :
- Characterizing production master data; (A1, A2, C1, C2, C6)
- Specifying a scheduling problem; (A1, A2, C1, C2, C3)
- Choosing or proposing a method to solve a production scheduling problem that is best suited to a given context, and applying it; (A2, A3, C2, C3, C5, C6)
- Analyzing the impacts of scheduling methods. (A1, A2, A3, C1, C3, C5, C6)

CONTENT

1- Lecture-tutorials part :
Introduction: Positioning scheduling within the overall production management process, characteristics of scheduling problems, problem classification.
Single-machine problems.
Parallel machine problems.
Flowshop problems.
Jobshop problems.
Hybrid problems.
2- Project: using a scheduling software package (Opcenter APS) to solve a case study
Study of production master data.
Analysis of software functionalities.
Consideration of additional constraints.
Software evaluation and critical analysis.

BIBLIOGRAPHY

BAGLIN G., BRUEL O., GARREAU A., GREIF M., Van DELF C., Management Industriel et Logistique, Economica, 1996.
GIARD V. Gestion de la production et des flux, 3e édition, Economica, 2003.
JAVEL G. Organisation et gestion de la production, Masson, 1997.
PINEDO M., Scheduling: Theory, Algorithms, and Systems, Prentice Hall, 2001.
RINNOOY KAN A.H.G., Machine scheduling problems: Classification, complexity and computations, Nijhoff, The Hague (1976).Hall, 2001

PRE-REQUISITES

Basics of Production Management (production organization, master data, production planning)
Basic knowledge about algorithms.